CSS RWD (Responsive Web Design) is a design approach that makes websites look and function well on all devices—desktops, tablets, and phones—using **flexible layouts**, **media queries**, and **fluid images**.

✅ Key Concepts of CSS RWD

1. Viewport Meta Tag

Ensures proper scaling on mobile devices:

```
<meta name="viewport" content="width=device-width, initial-scale=1.0">
```

---

2. Fluid Layout (Percent-based Widths)

Use percentages instead of fixed widths:

```
.container {
  width: 100%;
  max-width: 1200px;
  margin: auto;
  padding: 0 20px;
}
```

---

3. Responsive Media (Images & Videos)

Make media scale with screen size:

```
img, video {
  max-width: 100%;
  height: auto;
}
```

---

4. Media Queries

Apply different styles at specific breakpoints:

```
/* Mobile First */
body {
  font-size: 16px;
}

@media (min-width: 768px) {
  body {
    font-size: 18px;
  }
}

@media (min-width: 1024px) {
  body {
    font-size: 20px;
  }
}
```

---

5. Flexible Grid Layouts

Use Flexbox or CSS Grid to adapt to screen sizes.

🟦 Flexbox Example:

```
.flex-container {
  display: flex;
  flex-wrap: wrap;
  gap: 1rem;
}

.flex-item {
  flex: 1 1 100%;
```

```
}

@media (min-width: 768px) {
  .flex-item {
    flex: 1 1 48%;
  }
}

@media (min-width: 1024px) {
  .flex-item {
    flex: 1 1 30%;
  }
}
```

🟦 **Grid Example:**

```
css
CopyEdit
.grid {
  display: grid;
  grid-template-columns: 1fr;
  gap: 1rem;
}

@media (min-width: 768px) {
  .grid {
    grid-template-columns: repeat(2, 1fr);
  }
}

@media (min-width: 1024px) {
  .grid {
    grid-template-columns: repeat(4, 1fr);
  }
}
```